

Australia is free of the world's worst animal diseases such as foot-and-mouth disease and avian influenza H5N1. Animal pests and diseases are a major threat to Australia's livestock and poultry industries and an outbreak could impact on our access to export markets and undermine livelihoods. Exotic wildlife diseases could also risk the health of our unique native animals and introduce zoonotic viruses that could affect humans. There is an ongoing need to [practice good biosecurity](#). Please report the first signs of disease.

Biosecurity means protecting the economy, environment, and the community from the negative impacts of pests, disease, weeds, and contaminants.

Biosecurity practices include: disinfecting, signage, maintaining boundary fences, checking for strays, restricting visitor and vehicle movements, ensuring all machinery brought onto the property is cleaned, good husbandry, ensuring purchases are from reliable sources, inspecting the flock or herd regularly, quarantining new stock.

For detailed information, visit the [Farm Biosecurity website](#).

Report suspected diseases

If you spot anything unusual in livestock or birds report the suspect disease by phoning the Exotic Animal Disease Hotline on 1800 675 888. This hotline number will put you in touch with your local department of primary industries or agriculture.

Outbreak

Information about current national eradication programs is available on the [Government's Outbreak website](#).

This website also provides information on how to prevent and prepare for an outbreak and what to do in the event of an outbreak.

Pests and Diseases of Animals

Understanding the risk of African horse sickness virus establishing and spreading in Australia

Background

African horse sickness virus (AHSV) is a virus spread by biting midges (*Culicoides* species). It affects equids (horses, donkeys and zebras) and is present across a large area of sub-Saharan Africa. This virus causes severe disease, with up to 90% of infected, unvaccinated horses dying.

In 2020, African horse sickness (AHS) was detected in Thailand and Malaysia. This was the first time it had been reported in southeast Asia. Due to the success of control and eradication efforts, Thailand and Malaysia are now recognised as free from AHSV by the World Organisation of Animal Health (WOAH). AHS is a nationally notifiable disease and must be reported to agricultural authorities. AHSV has never been detected in Australia. If it entered Australia, it could be detrimental to the thoroughbred racing industry and all equestrian sectors.

There are approximately 300 species of *Culicoides* midges in the Australasian region, though it's not known whether these species could spread AHSV.

The study

To improve our understanding of this risk, the department commissioned CSIRO to analyse Australian biting midge species that feed on horses and assess their capacity to spread the virus.

The project team analysed midges collected through the National Arbovirus Monitoring Program. To identify potential horse-feeding species, the midge samples were tested for horse DNA.

The project team also conducted infection trials at CSIRO's Australian Centre for Disease Preparedness (ACDP) in Geelong. This is a high biocontainment site, designed to safely allow research into infectious diseases. These infection trials aimed to determine whether biting midge species - *C. brevitarsis*, *C. actoni* and *C. marski* - fed with virus-infected blood could become sufficiently infected to spread AHSV.

The results

The study identifying potential horse-feeding midge species detected small amounts of horse DNA in only two midge species - *C. bundyensis* and *C. austropalpalis*. This indicates these insect species feed on horses. However, due to the low number of midges collected, further research is needed to better understand potential horse-feeding Australian midge species. This may involve expanding the collection areas for midge specimens. More specimens could also be collected closer to areas where horses live.

The infection trials showed that none of the three biting midge species tested were positive for AHSV at the end of the testing period. More research would be required to confirm the results of this study. However, these initial findings suggest these three Australian midge species may not be capable of spreading AHSV. This doesn't rule out the competence of other, untested midge species to spread AHSV in Australia.

This project also improved our knowledge of the conditions required for the survival of wild-caught *Culicoides* during transport and in a laboratory. This information will be valuable for future research studies on *Culicoides* midges in Australia.

The project team would like to thank the National Arbovirus Monitoring Program for providing midge samples, and external collaborators for facilitating access to field sites.

ASF is a highly contagious viral disease of domestic and wild pigs. It is clinically similar to classical swine fever but it is caused by a different virus.

ASF is spread within domestic pig populations primarily through contact with infected pigs, movement of people or objects that have been in contact with infected pigs and feeding of contaminated swill. The virus can survive for long periods in uncooked, frozen or cured pig-meat products. It persists in contaminated pig pens for at least 30 days and is readily carried on equipment, shoes, clothing or vehicles. No effective treatment or vaccine is available for ASF.

The disease was first documented in sub-Saharan Africa in 1921 where it continues to be endemic. In 2007 ASF was reported in Georgia and it then spread into Russia, Ukraine and Belarus. In 2014 ASF first entered the EU in Lithuania and has since been detected in 11 member states extending as far west as Belgium. In August 2018, ASF was reported in China. It has since spread in order of occurrence to Mongolia, Vietnam, Cambodia, Hong Kong, North Korea, Laos and Myanmar.

An outbreak of ASF in Australia would have significant impacts on pig production, pig health and pig-meat export.

For more information visit the department's website pages on [keeping ASF out of Australia](#) and the [emergency animal disease bulletin](#). The [AUSVETPLAN disease strategy](#) for ASF is located on the Animal Health Australia website. General information on ASF can be obtained from the [Food and Agriculture Organization's \(FAO\)](#) or the [World Organisation for Animal Health's \(OIE\)](#) websites. Official information on the status and dates of ASF outbreaks can be obtained from the OIE's [World Animal Health Information System \(WAHIS\)](#).

Anthrax is an acute disease caused by infection with the spore-forming bacterium called *Bacillus anthracis*. It can affect a wide variety of domestic and wild animals. The disease usually occurs very suddenly in cattle and sheep. Affected animals are found dead often with no previous signs of illness. Humans are also susceptible to anthrax, although human cases are very rare in Australia. The risk of human infection occurs when examining, skinning or cutting up an infected carcass. The NSW Department of Primary Industries provides an [online fact sheet](#) with images of affected animals and other information about anthrax in livestock.

For more information about anthrax, consult the nationally agreed response plan [AUSVETPLAN Disease Strategy–Anthrax](#).

Australian bat lyssavirus is a virus of the family Rhabdoviridae that is closely related, but different, to classical rabies virus.

All lyssaviruses cause invariably fatal encephalitis (inflammation of the brain) in mammals. The natural history of ABLV is similar to rabies.

The virus is shed in the saliva, and it is presumed that ABLV is usually transmitted by bats through bites or contamination of a fresh wound, scratch, or mucous membranes with infected saliva.

For more information visit the [Australian bat lyssavirus page](#) on the Queensland DAFF website or consult the nationally agreed response plan AUSVETPLAN Disease Strategy—Australian Bat Lyssavirus.

Bird flu is a highly infectious disease of birds and some strains may affect humans. All bird species are thought to be susceptible to bird flu and the disease can affect more than 140 bird species. Many wild birds and waterfowl (especially geese, ducks and swans) carry the virus but generally don't show signs of the disease. However, they can infect other birds and poultry they come into contact with. For more information visit the department's website pages on [avian influenza](#) and [bird biosecurity](#) or consult the nationally agreed response plan AUSVETPLAN Disease Strategy—Avian influenza. More information about the disease is also available on the [Northern Australia Quarantine Strategy](#) section of this website, as avian influenza is a [target disease](#) under this strategy.

For information on pests and diseases of bees, or the response to [Asian honey bees](#) in Australia visit the department's [Bee Pests and Diseases](#) page. There is also good information available on the [BeeAware website](#).

Bovine spongiform encephalopathy (BSE) is a progressive, fatal disease that affects the nervous system of cattle. BSE is one of a number of transmissible spongiform encephalopathies (TSEs). Transmissible spongiform encephalopathies (TSEs) in animals are a class of rare brain diseases that are associated with the accumulation of abnormal prion protein in the brain which affects the central nervous system. These diseases are very rare, fatal and are characterized by spongy degeneration of the brain. There are no validated live animal tests, no treatments and no vaccines for these diseases. There are a number of TSEs which affect people and animals.

Further information about BSE, TSEs and the [Transmissible Spongiform Encephalopathy Freedom Assurance Program](#) is available on the Animal Health Australia website. For information about the nationally agreed response plan for BSE consult the AUSVETPLAN Disease Strategy—BSE.

CSF, also known as hog cholera or swine fever, is a highly contagious viral disease of pigs. It is clinically similar to African swine fever, but is caused by a different virus.

There have been several outbreaks in Australia that have been eradicated in the past. The last was in 1961. While infected pigs spread the disease the virus can also survive for a long time in frozen pig carcasses and cured and salted pork. It can remain infectious in contaminated pig pens for up to two weeks and can be carried on clothing, shoes or vehicles.

More information about the disease is available in the relevant [AUSVETPLAN manual](#) and on the [Northern Australia Quarantine Strategy](#) section of this website, as CSF is a [target disease](#) under this strategy.

A CSF outbreak would have serious consequences for Australia's domestic and export production of pigmeat. For more information about CSF consult the AUSVETPLAN Disease Strategy—CSF.

EBL is a disease of adult cattle due to infection with the bovine leucosis virus. Cattle may be infected at any age. Most infections do not show clinical signs. Some cattle over three years old develop persistent lymphocytosis and a smaller proportion develop lymphosarcomas (tumours) in various internal organs

leading to premature death. More information about EBL can be found on the [World Organisation for Animal Health website](#).

A scheme to eradicate EBL from the Australian dairy herd began in the mid 1990s and, based on herd testing results, freedom was achieved in the dairy industry on 31 December 2012. Following this achievement a standard (serving as the basis for the dairy industry to maintain freedom from the disease) was drafted (see the word document below). The 2012 Standard Definitions and Rules for Control and Eradication of Enzootic Bovine Leucosis incorporates key elements of the OIE requirements for EBL eradication.

Equine influenza (EI) is an acute, highly contagious, viral disease which can cause rapidly spreading outbreaks of respiratory disease in horses, donkeys, mules and other equine species. EI would have a major impact on the Australian horse industry if it were to become established here. For further information about signs of disease in horses see the [NSW Department of Primary Industries website](#).

An outbreak of EI occurred in the Australian states of New South Wales and Queensland in late 2007. [Eradication](#) was achieved in Australia in early 2008 by implementing a number of measures in accordance to the nationally agreed [AUSVETPLAN disease response manual](#). For more information on the outbreak and its eradication visit the [NSW Department of Primary Industries website](#).

An independent inquiry into the outbreak of equine influenza in Australia was completed in 2008. That report and the government response is available on the [department's website](#). A copy of the dossier is also available for [download below](#).

Despite EI having been eradicated from the Australian horse population, horse owners and handlers are strongly encouraged to stay vigilant and maintain good biosecurity.

Australia's freedom from equine influenza

On 25 December 2008, Australia self-declared freedom from EI in accordance with the World Organisation for Animal Health international standards. Evidence to support Australia's return to freedom was published in the below dossier.

[Foot-and-mouth disease](#) (FMD) is a highly contagious animal disease that would have severe consequences for Australia's animal health and trade. Australia has detailed FMD [response plans and arrangements](#) in place, while government and industry preparedness is under continuous [review](#).

Improvements to the national capability are constantly implemented and exercises are held regularly to test plans and train those who would be involved. [Research](#) and reviews of new technology relating to FMD control are also undertaken. More information about [FMD](#) is available on the [Northern Australia Quarantine Strategy](#) section of this website, as FMD is a [target disease](#) under this strategy. An [AUSVETPLAN Disease Strategy manual for FMD](#) is available on the [AHA website](#).

Australia also has a [national vaccination policy](#)

[Hendra virus](#) is a zoonotic disease, which means it can transfer from animals to people. Hendra virus can cause disease in horses but only rarely in humans. It can be transmitted from flying fox to horse, horse to horse, and horse to human. Read more about signs of Hendra virus in horses. For signs of Hendra virus in people, see the [Queensland Health website](#). The first reported case of a dog testing positive for the Hendra virus outside of laboratory conditions was announced on 27 July 2011. It is recommended that people keep companion animals, such as [dogs](#) and cats, away from sick horses to reduce the risk of an infection as well as taking other [steps to reduce the risk of becoming infected with Hendra virus](#). Steps can also be taken to reduce the risk of [horses](#) becoming infected with Hendra virus, including biosecurity measures.

For more information see the AUSVETPLAN response policy brief for Hendra virus infection.

A Hendra virus vaccine ‘Equivac HeV’ is available through veterinary practices in Australia. For more information about the vaccine visit the Australian Veterinary Association’s webpage or the [NSW Government website](#).

The Australian Government Department of Agriculture, Fisheries and Forestry has a role in:

- managing market access issues that may arise
- analysing and sharing technical information with other government departments (such as the Department of Health)
- making sure all states and territories are provided with situation updates by the affected states
- international notification through the World Organisation for Animal Health of significant developments.

Infectious bursal disease (IBD) is a highly contagious viral disease of domestic fowl. The virus can remain infective in the environment for long periods and is resistant to many disinfectants.

Highly virulent strains of IBD virus have emerged in Europe, Asia and the United States over recent years. Low-pathogenicity strains of IBD virus occur in Australian poultry flocks, but there is no evidence of highly virulent strains here.

Illness usually occurs in young birds aged three to six weeks. Infected birds pick at their vents and are reluctant to move.

Other signs are not specific but include depression, not eating, ruffled feathers, trembling, watery (sometimes bloody) diarrhoea, and sudden death, usually within four days of signs appearing.

Highly virulent IBD is a serious threat to production in commercial poultry flocks. Highly virulent strains can cause mortalities of up to 25 per cent in broilers and up to 60 per cent in laying pullets. More information about virulent strains is available in the relevant [AUSVETPLAN manual](#).

[Japanese encephalitis](#) is a viral zoonotic disease that is spread by mosquitoes. The virus can cause reproductive losses and encephalitis in pigs and horses. In rare cases, Japanese encephalitis can cause

disease in people. People and horses are considered 'dead end' hosts. Once infected, they do not play a role in transmitting the virus. Pigs and some species of wild birds are amplifying hosts.

Johne's disease, otherwise known as paratuberculosis, is a chronic wasting disease of ruminants (e.g. cattle and sheep).

Johne's disease is caused by the bacteria *Mycobacterium avium* subsp. *paratuberculosis* (M. *paratuberculosis*). Different strains of the bacteria usually affect different animals:

- Ovine Johne's disease (OJD), caused by ovine strains of M. *paratuberculosis*, affects mainly sheep and goats.
- Bovine Johne's disease (BJD), caused by bovine strains of M. *paratuberculosis*, affects mainly cattle, goats, deer and alpaca.

Both strains are endemic in Australia, although the disease is not common.

The disease can have serious economic effects due to production losses if it is not controlled. More information about both diseases is available on the [Animal Health Australia website](#).

A revised national approach to the management of Johne's disease in cattle was agreed by key stakeholders (including Australian livestock industries, government and the veterinary profession) and implemented in 2016. Details about the approach can be found at the [Animal Health Australia website](#).

Lumpy skin disease (LSD) is a serious disease of cattle and water buffalo. It has never occurred in Australia but is an emerging threat as it continues to spread through Asia, including Indonesia. Border requirements are in place for incoming air and sea passengers, imported cargo and mail items to ensure biosecurity risks are managed. Returning livestock vessels are also managed by biosecurity officers. The [AUSVETPLAN for lumpy skin disease](#) is part of our national emergency response arrangements. It sets out the nationally agreed approach that would be taken to respond to lumpy skin disease if it occurred in Australia.

ND is a viral disease of domestic poultry and wild birds, which is characterised by gastrointestinal, respiratory and nervous signs. Human infection with ND virus is extremely rare and usually occurs only in people who have close direct contact with infected birds – for example, poultry processing workers, veterinarians or laboratory staff. The virus causes only mild, short-term conjunctivitis or influenza-like symptoms in humans.

Australia is free from virulent ND but has contained and eradicated [a number of outbreaks](#):

- September 1998 - western Sydney and Rylstone, New South Wales
- April 1999 - Mangrove Mountain, New South Wales
- August 1999 - western Sydney, New South Wales
- January/February 2000 - western Sydney and Tamworth, New South Wales
- May 2002 - Meredith, Victoria
- October 2002 - western Sydney, New South Wales.

Australia has a [Newcastle Disease Management Plan](#), an [AUSVETPLAN](#) disease strategy and [vaccination](#) is widespread in commercial poultry flocks.

More information about the disease is available on the [Northern Australia Quarantine Strategy](#) section of this website, as ND is a [target disease](#) under this strategy.

Rabies is a fatal viral disease of warm-blooded animals, including humans. It is usually spread by the bite of an infected animal such as a dog, cat, bat or fox. Rabies is present in many countries in the world, except Australia, New Zealand, the United Kingdom, Japan and a number of small islands. Many of these are island nations with strict controls on the entry of animals. Both of these factors have helped prevent entry of the disease. More information about rabies is available on the [Northern Australia Quarantine Strategy](#) section of this website as rabies is a [target disease](#) under this strategy.

More information about rabies is available in the relevant [AUSVETPLAN manual](#).

For human health information about rabies visit the [Department of Health website](#).

Screw-worm fly is an insect parasite of warm-blooded animals, including humans. Related to the blowflies that cause fly-strike in Australian sheep, it prefers hot, humid climates and cannot survive in frost-prone areas.

There are two species of screw-worm fly: 'Old World' (*Chrysomya bezziana*) and 'New World' (*Cochliomyia hominivorax*). More information about screw worm fly is available in the relevant [AUSVETPLAN manual](#)

Screw-worm flies could cost close to \$500 million a year in lost production and control measures if they entered Australia. They would have a devastating effect on northern livestock production, particularly cattle and sheep industries.

In some herds, 10 to 15 percent of cattle could be struck at any time; the greatest loss would be through the deaths of newborn calves as a result of navel strike.

The potential spread of this pest from Papua New Guinea is one of the major biosecurity threats to northern Australia. Eradication would depend on the release of millions of sterile male screw-worm flies to reduce breeding numbers. This could take up to five years, with health implications for remote northern communities.

More information about [old-world screw-worm fly](#) is available on the [Northern Australia Quarantine Strategy](#) section of this website, as screw-worm fly is a [target disease](#) under this strategy.

The Department of Department of Agriculture, Fisheries and Forestry has a monitoring program in the Torres Strait, on Cape York Peninsula and in other strategic locations to ensure early detection of a screw-worm fly incursion. The program includes lure traps for adult screw-worm flies around airports, seaports and other high-risk areas, and inspection of livestock for fly strike. Animal health personnel also inspect returning livestock vessels, and sentinel cattle herds are used to monitor for maggots.

Surra is a chronic wasting disease caused by a species of trypanosomes. The disease, which is spread by biting flies, is most severe in horses, donkeys, mules, deer, camels, llamas, dogs and cats, but also

occurs in cattle, buffaloes, sheep, goats, pigs and elephants. More information about surra is available in the relevant [AUSVETPLAN manual](#).

Surra would be difficult or impossible to eradicate if it were to become established in Australia because it could persist in feral animals such as horses, donkeys, deer, camels, buffaloes, goats and pigs. Control of the disease in livestock could cost cattle, sheep and horse industries millions of dollars.

Scientific studies have demonstrated that kangaroos, agile wallabies and pademelons are highly susceptible to infection with surra, which in experimental conditions causes acute clinical signs with high mortality. These findings suggest that surra could also have a significant effect on some native Australian marsupials.

More information about surra is available on the [Northern Australia Quarantine Strategy](#) section of this website, as surra is a [target disease](#) under this strategy.

Pests and diseases that must be reported

- [National List of Notifiable Animal Diseases](#)
- [National List of Reportable Diseases of Aquatic Animals](#)
- [\(CCIMPE\) Trigger List for Introduced Marine Species](#)
- [State and Territory Notifiable Animal Diseases lists](#)

Source: <https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/animal>